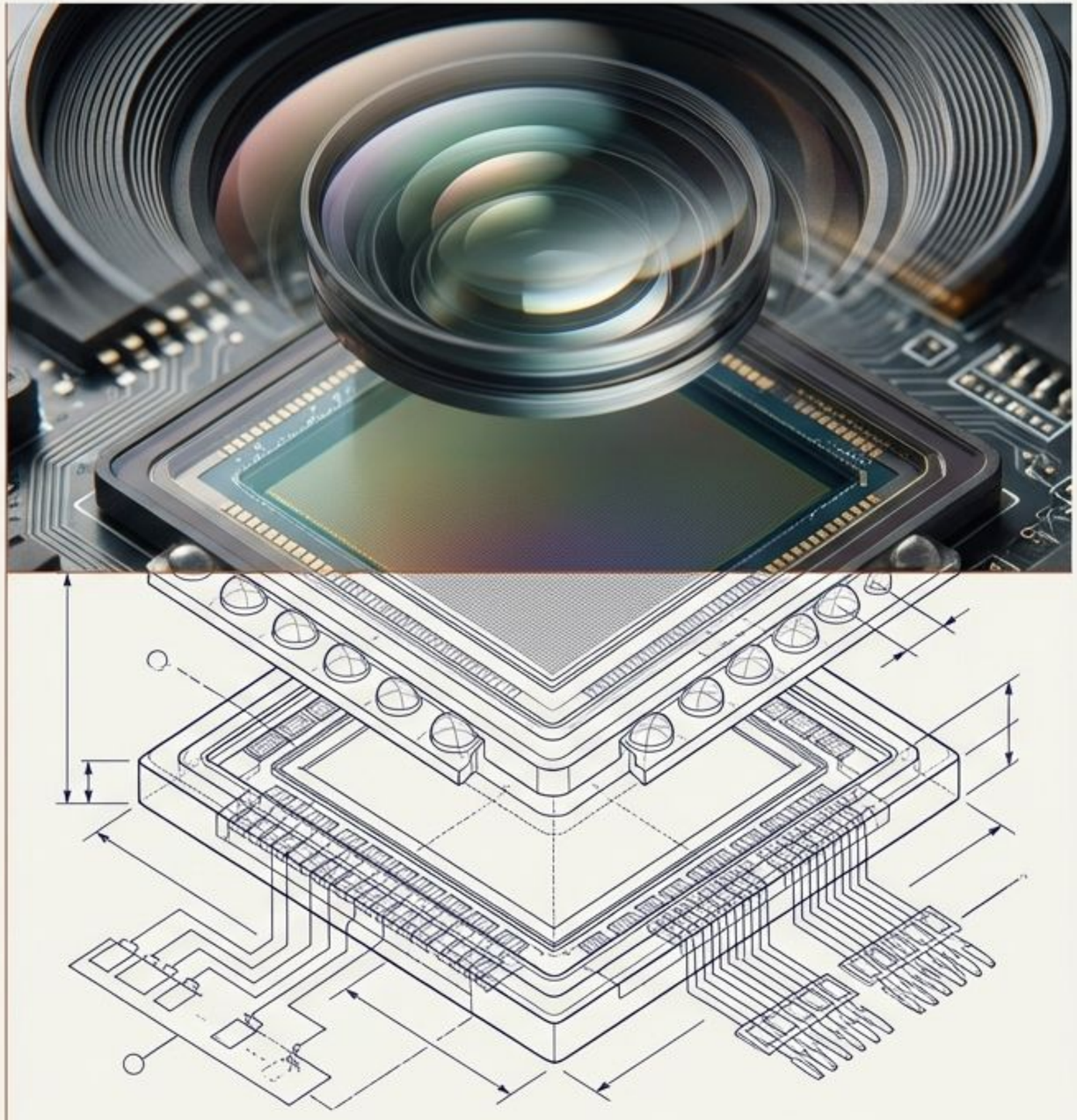
Images and Graphics in Multimedia

Module 3: Visual Architecture and Data Lifecycle



Prepared By : Er. Sharat Maharjan

The Real-World Impact of Visuals

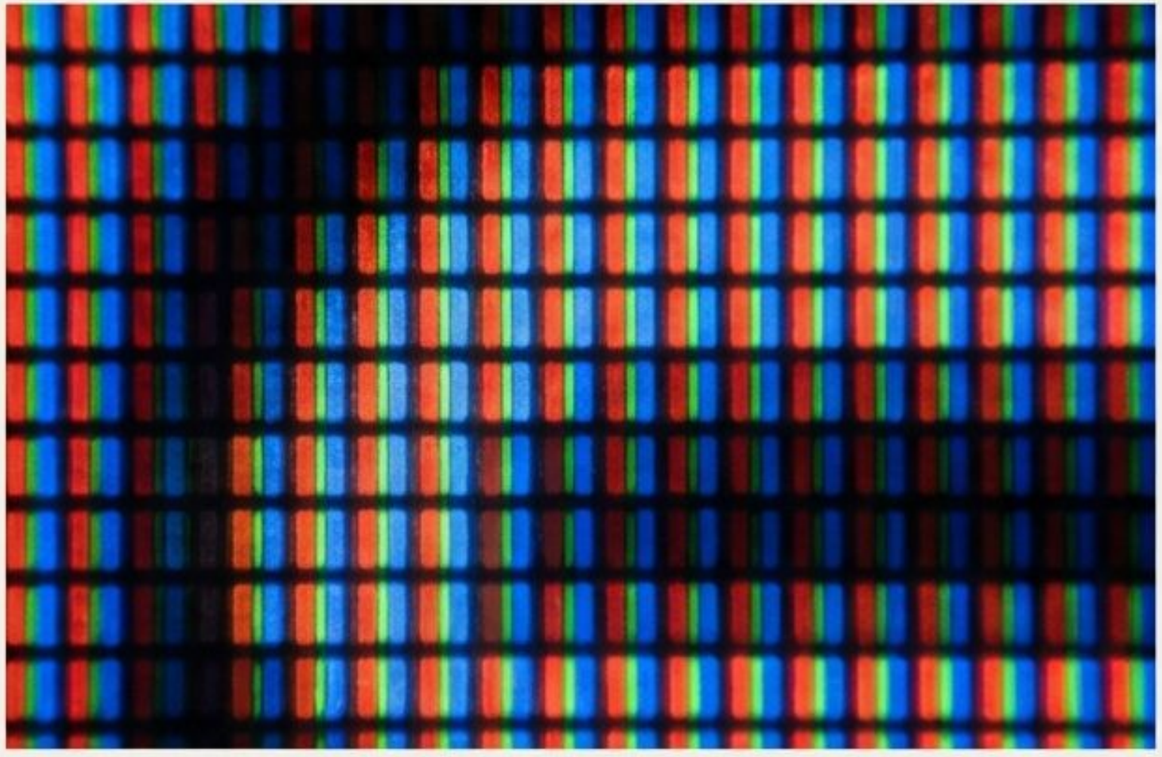
Visual Appeal Engages users in mobile and game interfaces.	**Fast Information** Brains process visuals faster than complex text.	**Learning & Retention** Diagrams improve cognitive memory.
UI Navigation Intuitive icons guide user actions.	**Brand Identity** Vector symbols create immediate recognition.	**Creative Expression** Digital painting and photo manipulation.

Two Pillars of Digital Representation

Computers process images through two primary architectural models, dictating file size, scalability, and optimal use.

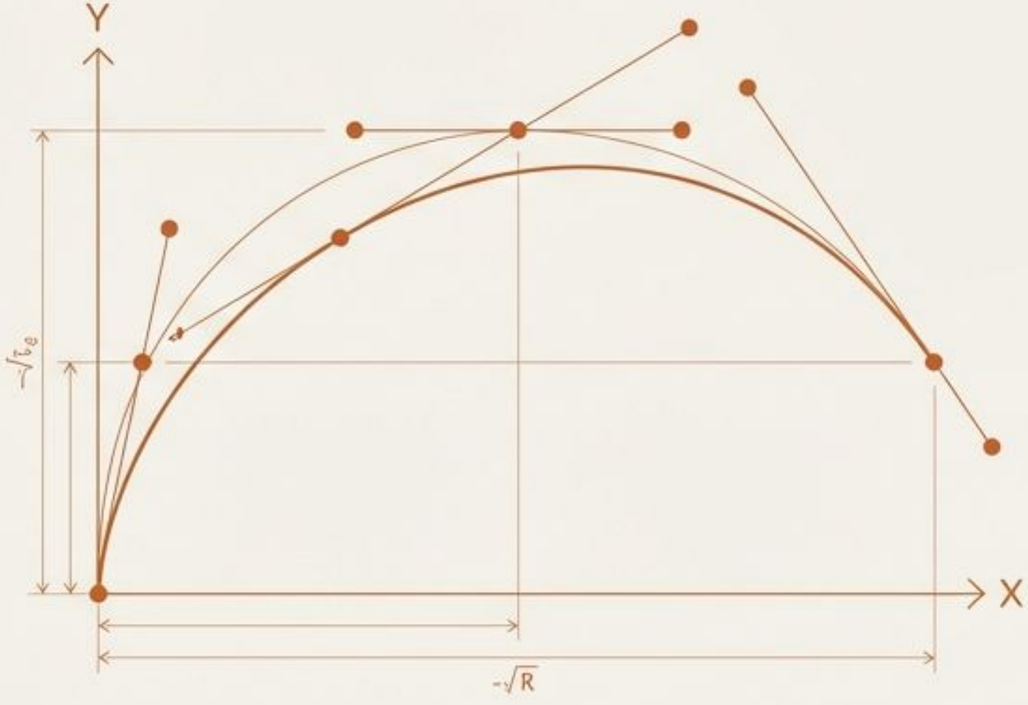
Raster (Bitmap)

Built on a structural grid.



Vector

Built on mathematical instructions.



Raster Graphics: The Pixel Matrix

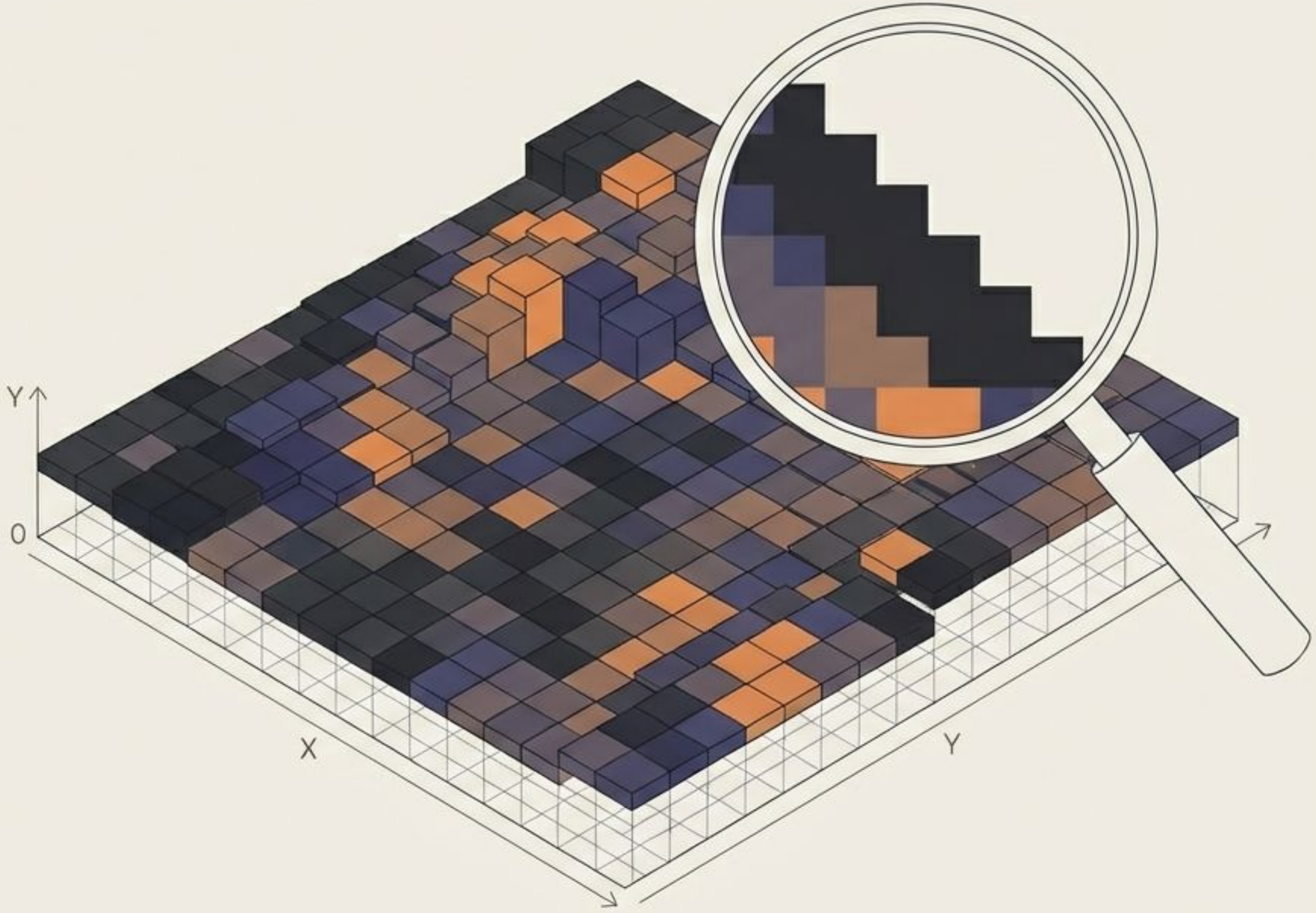
Composition
A 2D rectangular array of pixels.

Data Structure
Each pixel stores specific color information.

Resolution-Dependent
Quality is fixed at capture.

Scaling Penalty
Enlarging causes visible pixelation and blur.

Optimal Use
Continuous tones and photographs (e.g., JPEG, TIFF).



Vector Graphics: The Mathematical Path

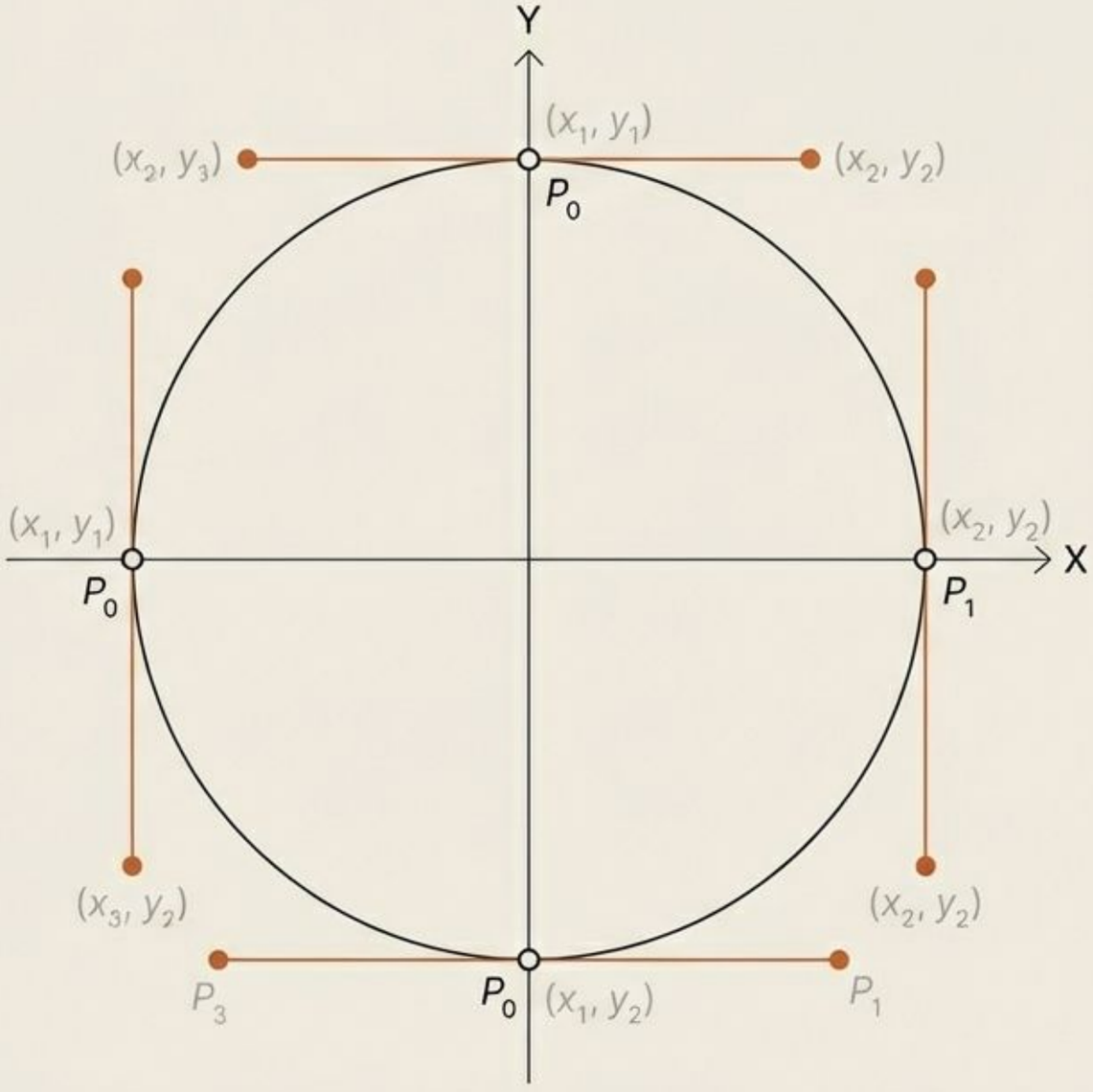
Composition
Defined by geometric primitives (points, lines, curves).

Data Structure
Stored as mathematical rendering instructions.

Resolution-Independent
Perfectly retains sharpness at any size.

Infinite Scaling
Enlarging does not degrade quality.

Optimal Use
Typography, logos, and blueprints (e.g., SVG, TrueType).



Head-to-Head: Raster vs. Vector

	Raster	Vector
Composition	Pixels (dots in a fixed grid)	Mathematical paths and curves
Scalability	Loses quality (pixelates upon zoom)	Infinitely scalable with perfect edges
File Size	Generally larger (stores every pixel)	Generally smaller (stores equations)
Real-World Use	Photographs and complex continuous tones	Logos, illustrations, and typography

Anatomy of a Pixel: Depth & Digitization

Resolution

Total pixel count (e.g., 1920x1080) defining detail level.

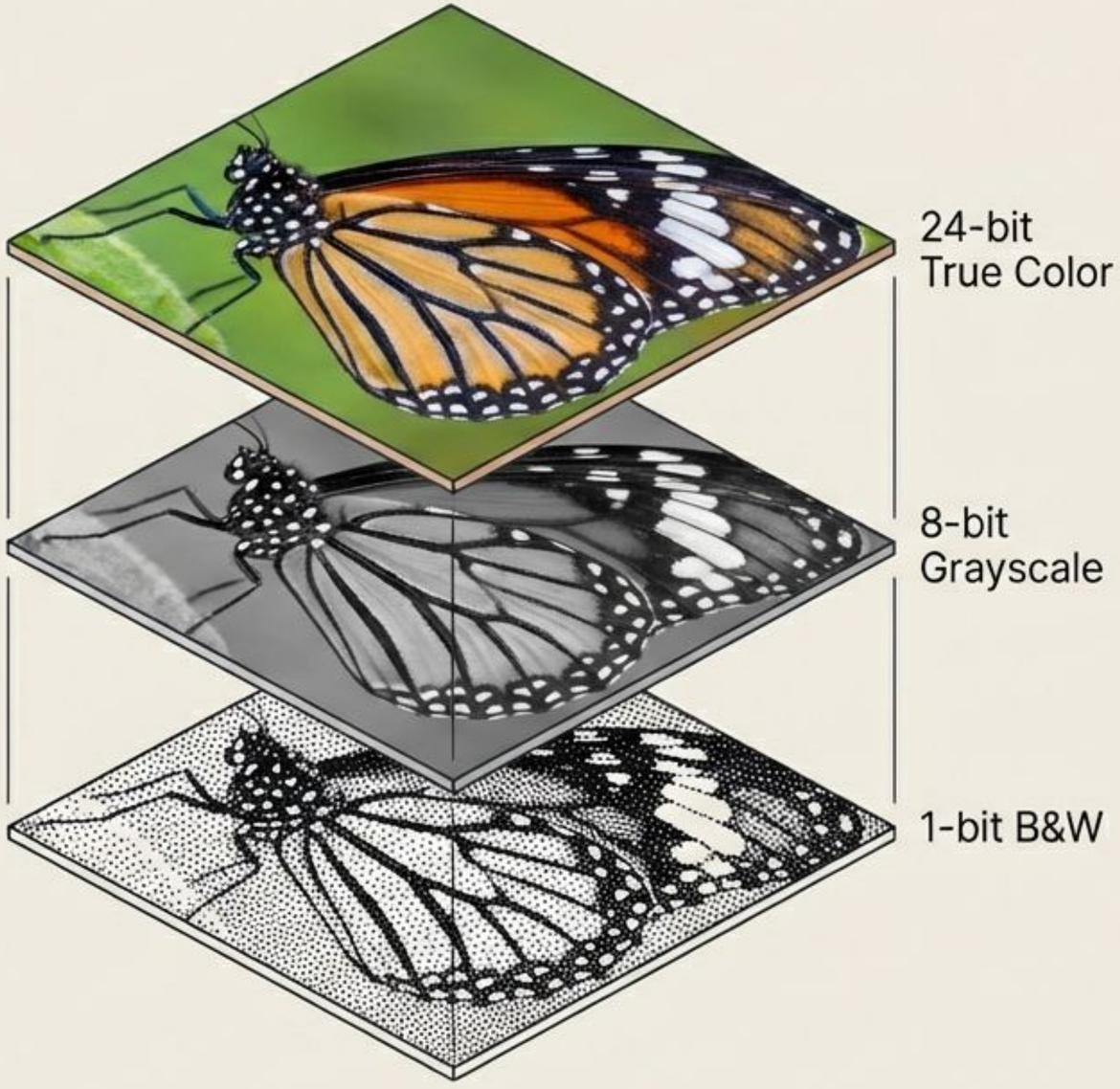
Bit Depth

Memory allocated per pixel for color.

- **1-bit**: Black & White
- **8-bit**: Grayscale (256 shades)
- **24-bit**: True Color (16.7M colors)

Digitization Process

- **Sampling**: Digitizes the coordinate grid (Resolution).
- **Quantization**: Digitizes the amplitude values (Bit Depth).



The Web Format Matrix

JPEG

24-bit RGB. Lossy compression. Ideal for standard web photographs.

PNG

24-bit & 8-bit. Lossless. Supports alpha transparency for logos and web graphics.

GIF

8-bit max (256 colors). Lossless (LZW). Supports basic animation and memes.

WebP

24-bit. Modern standard offering both lossy and lossless compression. Smaller files with transparency.

The Professional & Vector Matrix

TIFF

Up to 48-bit. Lossless. Preserves layers. Standard for print and professional publishing.

RAW

Very high bit-depth. Unprocessed sensor data. Used exclusively in professional camera workflows.

SVG

XML-based. Lossless text format. Standard for infinitely scalable web logos and icons.

BMP

Legacy Windows format. Uncompressed or RLE lossless. Generates massive file sizes, rarely used on the web.

The Multimedia Creator's Toolkit

Raster Editors (Paint)

Photo editing and digital art. (Photoshop, GIMP)
Inter Regular

Vector Editors (Draw)

Logos, layouts, and typography. (Illustrator, CorelDRAW)
Inter Regular

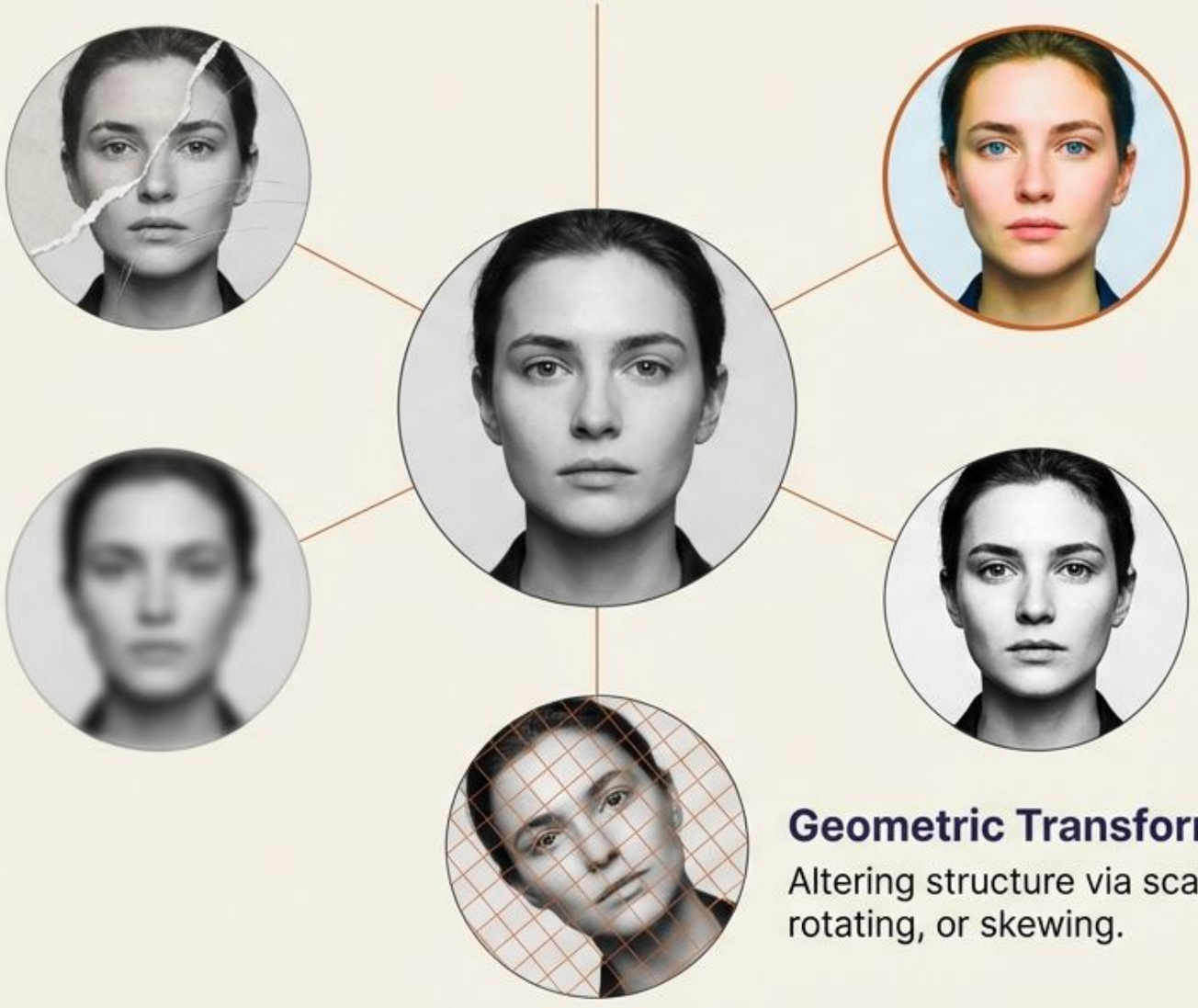
CAD Software

Precision technical drawing and engineering. (AutoCAD, SketchUp)
Inter Regular

3D Graphics

Modeling, texturing, rendering, and animation. (Blender, Maya)
Inter Regular

Image Manipulation Techniques



Restoration
Repairing physical damage in digitized heritage media.

Color Adjustment
Mathematically shifting hue, saturation, and exposure.

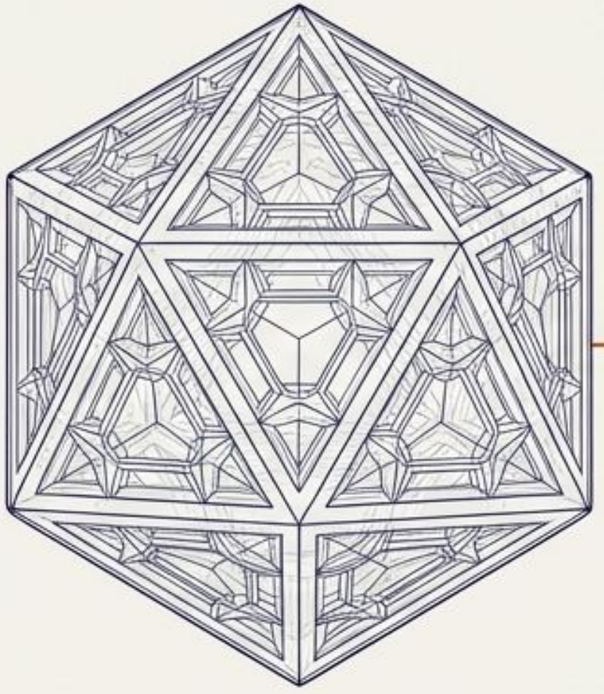
Filtering
Applying algorithmic effects like blur or edge detection.

Enhancement
Improving quality via brightness and contrast.

Geometric Transformation
Altering structure via scaling, rotating, or skewing.

Image Synthesis: Algorithmic Creation

Generating visual data from scratch through mathematical instructions.



Procedural Generation

Using mathematical formulas to auto-generate underlying geometry.

Digital Painting / Surfacing

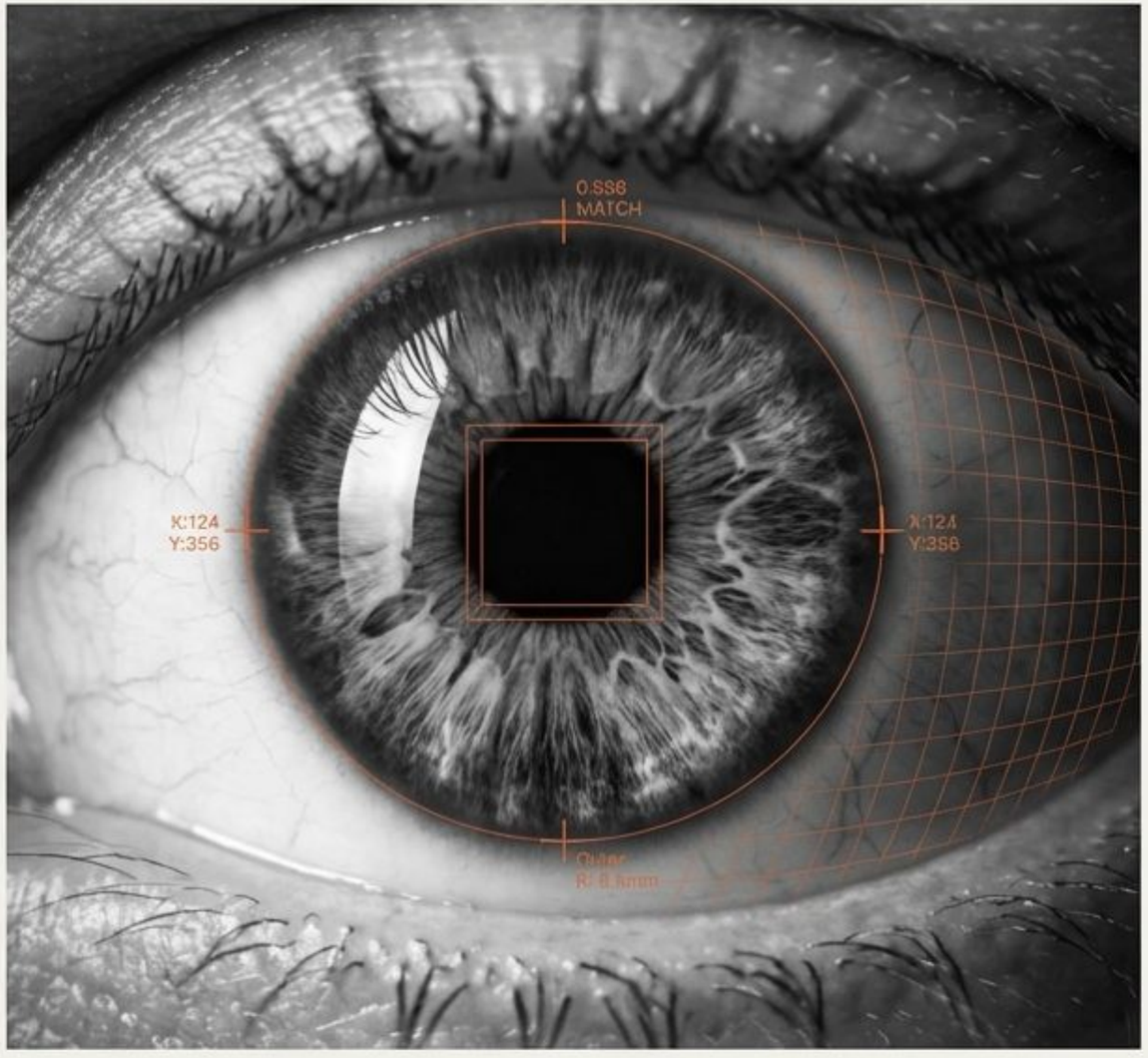
Applying manual textures, mapping, and base material properties.

Rendering

Calculating light bounces and texture to turn 3D scenes into 2D pixel frames.

Image Analysis: Computer Vision

Extracting meaningful data and semantic understanding from raw pixels.



Key Processes

- Segmentation: Partitioning into regions.
- Feature Extraction: Finding edges and corners.
- Pattern Recognition: Matching data sets.

Applications

- Facial Recognition: Biometric security.
- Medical Diagnostics: Detection of anomalies in MRI/CT.
- OCR: Optical Character Recognition for documents.

Image Transmission Over Networks

- **Bandwidth Challenges**
  High-resolution graphics demand significant network capacity.
- **Compression**
  Essential prerequisite for reducing file payload before delivery.
- **Progressive Transmission**
  Sending data in multiple passes. Displays a low-quality preview instantly, refining detail over time.
- **Quality of Service (QoS)**
  Protocols managing bandwidth and preventing packet corruption.

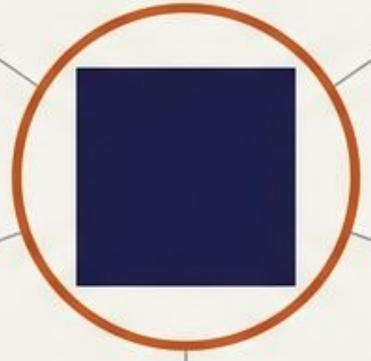
Module Summary & Key Takeaways

Impact

Visuals reduce cognitive load and drive UX.

Digitization

Resolution sets detail; Bit Depth sets color capacity.



Architecture

Raster (pixels, fixed grid) vs. Vector (math, infinite scale).

Formats

Chosen based on compression (Lossy/Lossless) and use case.

Lifecycle

Synthesis creates, Analysis understands, Transmission delivers.